

Porous Liquids: Computational Design for Targeted Gas Adsorption

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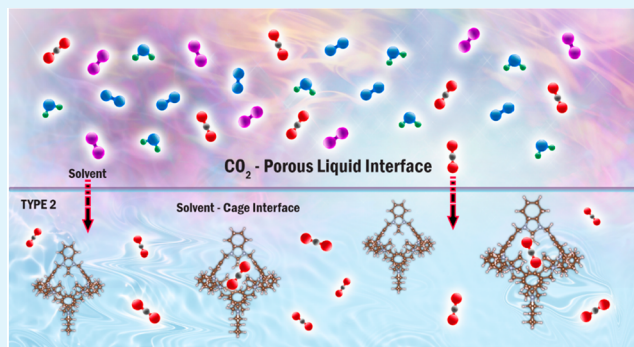
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ABSTRACT: In this *Perspective*, we present the unique gas adsorption capabilities of porous liquids (PLs) and the value of complex computational methods in the design of PL compositions. Traditionally, liquids only contain transient pore space between molecules that limit long-term gas capture. However, PLs are stable fluids that contain permanent porosity due to the combination of a rigid porous host structure and a solvent. PLs exhibit remarkable adsorption and separation properties, including increased solubility and selectivity. The unique gas adsorption properties of PLs are based on their structure, which exhibits multiple gas binding sites in the pore and on the cage surface, varying binding mechanisms including hydrogen-bonding and π - π interactions, and selective diffusion in the solvent. Tunable PL compositions will require fundamental investigations of competitive gas binding mechanisms, thermal effects on binding site stability, and the role of nanoconfinement on gas and solvent diffusion that can be accelerated through molecular modeling. With these new insights PLs promise to be an exceptional material class with tunable properties for targeted gas adsorption.

KEYWORDS: porous liquids, gas adsorption, gas separation, molecular modeling, material design



INTRODUCTION

Efficient energy production is one of the paramount challenges of modern civilization. Past reliance on fossil fuels such as coal and natural gas has resulted in a critical need for highly tunable and selective materials for separation and adsorption of greenhouse gases. Most notably, CO₂ removal is necessary in gaseous mixtures ranging from postcombustion streams to ambient air.^{1,2} To answer this challenge, multiple material and liquid systems are being investigated for their gas adsorption and separation properties, including metal–organic frameworks (MOFs), zeolites, porous organic cages (POCs), aqueous amines, and ionic liquids.^{3–7} Liquid amine-based separation systems have been the most extensively applied for commercial CO₂ capture, since they form stable carbonate-based intermediates that can be regenerated at elevated temperatures.^{8,9} These liquid systems are volatile and humidity sensitive, which limits their use for CO₂ capture, particularly in dilute conditions.¹⁰ Alternatively, porous materials have high selectivity and uptake adsorption capacity that is tunable via surface functionalization which, along with high mechanical stability, makes them extremely attractive for gas separation.^{11,12} The high adsorption capacity and strong gas binding via chemisorption creates challenges for regeneration, and the material synthesis is difficult to upscale, limiting the use of these materials for large-scale carbon capture.¹¹ Ultimately, there is a clear technological demand for regenerable material

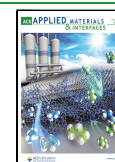
systems that enable selective and high-capacity CO₂ adsorption.

To answer these challenges a new class of porous materials was developed in the early 2000s that combines the permanent and rigid porosity from porous solids with the heat/mass transfer and fluidity of liquids. These new materials have been termed *porous liquids* (PLs). PLs are stable fluids that contain permanent porosity due to the combination of a rigid host structure and a solvent that is sterically excluded from entering the internal pore space. Many excellent recent review articles on PLs^{13–20} provide in-depth discussions on related PL topics such as formation, structure, and properties. In this *Perspective*, we present an overview of the state of PLs and their role in transformative technology for CO₂ capture, along with our views of challenges and opportunities in this field for the future. In particular, we will discuss the opportunities for rapid material discovery via modeling and simulation approaches to fine-tune PL structures for gas adsorption.

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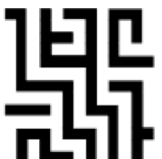
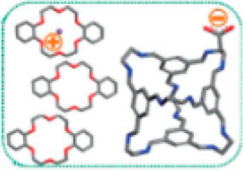
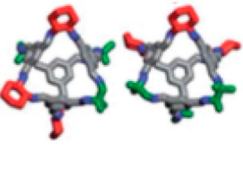
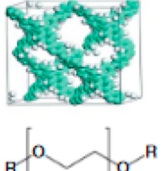
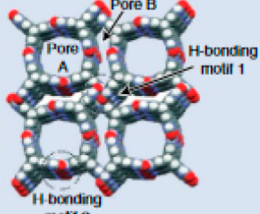
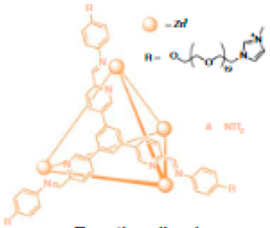
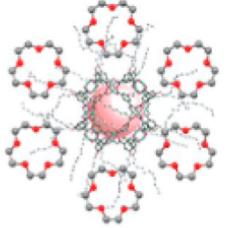
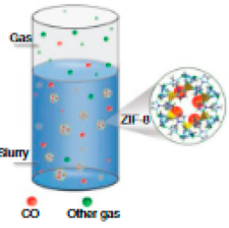
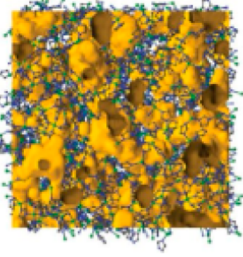
		Type			
		1	2	3	4
	Inorganic	 Functionalized SiO ₂	 Functionalized Mo ₁₅₄ rings	 Hydrophobic hollow SiO ₂ in water	 Mixed Si/Ge zeolites
	Organic	 Crown ether anionic covalent cages	 Scrambled organic cages	 PAFs in oil	 Triptycene trisbenzimidazole (or other HOFs)
	Inorganic-organic	 Functionalized metal-organic polyhedra	 Metal-organic polyhedra in crown ether	 ZIF-8 in glycol-2-methylimidazole	 Liquid ZIF-4

Figure 1. First row are schematics of PL types. Type 1 PLs consist of intrinsically porous hosts (open circles). Type 2 PLs contain dissolved porous hosts in sterically hindered solvents (filled circles). Type 3 PLs consist of microporous hosts (square grids) within a sterically hindered solvent. Type 4 PLs are neat microporous hosts that form transient, strongly associated liquids. Examples of known materials are given, alongside suggestions (shaded in blue) for some categories not yet populated. Reproduced with permission from ref 27 (2021, Springer Nature).

OVERVIEW OF POROUS LIQUIDS (PLS)

PLs consist of a thermodynamically stable combination of a solvent and a porous host that forms a liquid phase with permanent porosity following straightforward synthesis methods such as solution blending or magnetic stirring.^{21–23} While neat liquids contain transient porosity due to cavities that exist between solvent molecules, PLs have permanent porosity that is available for gas adsorption due to steric exclusion of the surrounding solvent from the internal pore space. PLs have been separated into four distinct types. Type 1 PLs consist of a neat liquid that is composed of molecules with stable intrinsic porosity, for example a liquid formed of hollow nanospheres.²⁴ Type 2 PLs consist of a rigid porous host and a solvent that is

sterically hindered from entering the pore cavity, for example, a porous organic cage molecule solvated in large crown-ether molecules.²⁵ Type 3 PLs consist of a solvated nanoporous particle, rather than distinct porous cages, in a sterically hindered solvent. For example, ZIF-8 nanoparticles solvated in a glycol-based solvent.²⁶ Finally, Type 4 PLs consist of a single component porous material that has a low melting point sufficient to form a liquid at room temperature. See Figure 1 for schematics of the four types of PLs and exemplar organic and inorganic compositions solvents and porous host material compositions.

Among these four types of PLs, Type 1 and Type 4 are the rarest, since they are composed of a single component that must be highly engineered to form a stable fluid. Type 2 PLs

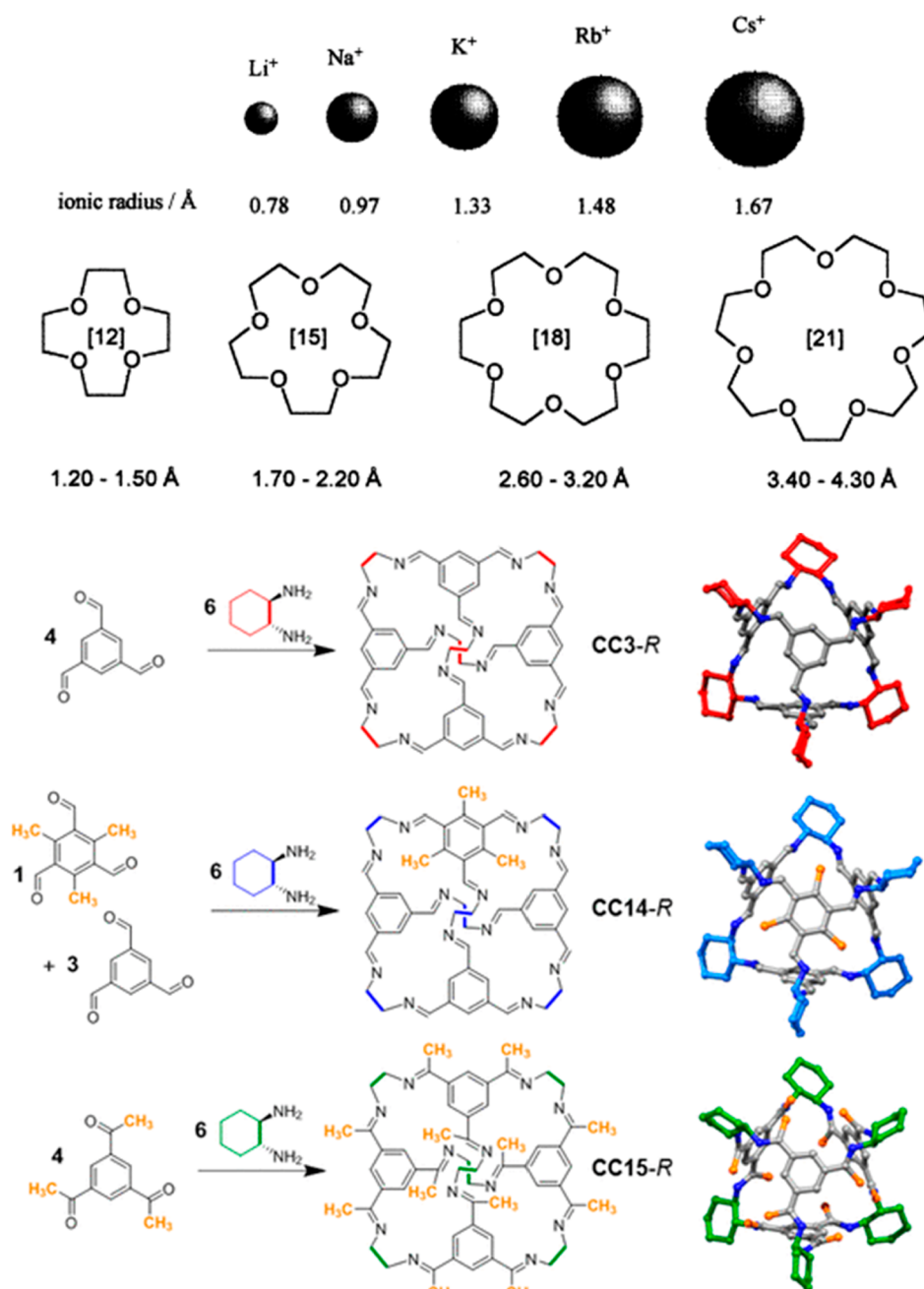


Figure 2. (top) Comparison of ionic radii of alkali metal cations with four different crown ether cavity diameters adapted from ref 39 (Elsevier, 2001) and (bottom) synthesis and schematic representation of POC cage molecules CC3-R, CC14-R, and CC15-R. In CC14-R three of the four cage windows are blocked by a single group, which in CC15-R all four cage windows are blocked by three methyl groups. Hydrogen atoms are omitted for clarity. Adapted from ref 40.

are true thermodynamic solutions and exhibit high porosity and stable liquid phase characteristics. Generally, Type 2 PLs have been the most extensively evaluated for their gas adsorption and separation capabilities. Type 3 PLs form suspensions or slurries and require control over particle size and solid–liquid interactions in order to form highly stable suspensions. While all four of the PL compositions exhibit distinct gas adsorption properties, the high tunability of the Type 2 and Type 3 PL compositions have the widest material design space for gas adsorption and separation.²³ Therefore, the remainder of this *Perspective* will focus primarily on the highly tunable Type 2 and Type 3 PLs, and how a

computational material design approach will be necessary for fine-tuning PL compositions and structure for optimized gas adsorption.

PL COMPOSITIONS

Type 2 and Type 3 PLs can potentially be fabricated from an almost limitless combination of inorganic or organic solvents and porous hosts. The most critical feature of a porous host material used in a Type 2 or Type 3 PL is the ability to dissolve without the loss of internal pore space.²⁸ Therefore, a stable and rigid pore structure is necessary, along with high solubility

and narrow pore apertures that sterically limit infiltration by a solvent molecule.

Generally, there are three different types of porous hosts that meet these requirements, POCs used in Type 2 PLs, and either MOFs or zeolites for Type 3 PLs. The first reported Type 2 PL was a tetrahedral POC functionalized with six crown ether substituents and a 15-crown-5 solvent reported by James et al. in 2015.¹⁵ Starting with this initial composition, multiple POC structures including CC3-R, CC14-R, and CC15-R POCs have been included in PL compositions, as seen in Figure 2. Additionally, significant progress been made through the development of “scrambled” cage POCs with varying functionalization that frustrates crystal packing and increases solubility.^{28–30}

MOFs are more commonly reported in Type 3 PL compositions, which form stable suspensions with rigid host structures. The most common MOFs used in PLs include ZIF-8 and HKUST-1.^{31–33} Functionalization of the MOF surface has been shown to enable PL formation through a compatibility principle, where similarities in surface groups on UiO-66 and in the coordinating solvent enables the formation of a stable suspension.³⁴ Along with MOFs, zeolites such as Silicalite-1 and ZSM-5 have also been successfully incorporated into Type 3 PLs.²²

Solvent selection in PL compositions alters the density, viscosity, solid dispersion, intermolecular interactions, and gas permeability of the resulting PL.³¹ The most critical features of a solvent in a PL is its size; it must be large enough to be hindered from entering the pore space. As a result, the earliest PL compositions used crown-ethers which are large cyclic solvents, as seen in Figure 2.¹⁵ Recently, investigations have focused on ionic liquids as the solvent in PLs due to their low vapor pressure, wide liquidus range, low viscosity, and high tunability through the addition of ions or organic functionalization.³⁵ Ionic liquids commonly used in PL compositions include [DBU-PEG][Tf₂N]₂, [P_{6,6,6,14}][NTf₂], and TOMAC.^{22,31,36–38} Cahir et al.²³ also identified that solvents such as silicone oils, polyethylene glycols, and naturally occurring triglyceride oils can all be used for PL formation, demonstrating the immense variability in possible PL compositions. Therefore, the challenge becomes not only to identify which solvents and porous hosts *can* be combined, but rather *which* porous host and solvent combination will produce a stable liquid with targeted gas adsorption properties.

To date, only a limited number of PLs compositions have been synthesized. PL synthesis methods are relatively straightforward, consisting of the combination of the solvent and porous host with a stirring and heating protocol.¹³ Therefore, the true synthesis challenge is in identifying the combination of porous host and solvents that form stable PLs with unique gas adsorption properties. Overall, the number of sterically hindered solvents and nanoporous materials that are available for PL formation is considerable. Among POCs, MOFs, and zeolites, more than 91 000 distinct cage structures have been reported.^{41–43} This number of possible porous structures does not account for functionalized subtypes that have been developed specifically for gas capture. For the solvent portion of the PL, the environmental protection agency reports ~4000 unique neat solvents along with an estimated 10¹⁸ possible ionic liquid compositions.⁴⁴ As a result, there are a minimum of 9 × 10²² possible combinations to test. Even if 99.9999% of these compositions are excluded based on a

combination of solvent and pore size, more than 10¹⁶ combinations would still need to be evaluated.

Therefore, identification of PL compositions and prediction of their gas adsorption and separation properties requires a more sophisticated scientific approach. Robotic discovery of new PL compositions was performed by Kearsey et al. and identified 29 new Type 2 PLs.²⁹ Despite the success of this high-through-put approach, machine learning and other data science approaches have been rarely applied to evaluation of PLs, particularly compared with the extensive evaluation of other porous materials.⁴⁵ Instead, the current focus is on the discovery of new POCs, which can then be applied to development of PLs.^{46–50} The result is increasingly large structural databases containing 60 000+ POCs and cage-within-a-cage structures that may be used in potential Type 2 PL compositions.^{47,51} Moreover, open source Python packages have been created to aid in structure prediction of POCs⁵² and inorganic cages.⁵³ This work on new cage structures will need to be expanded and combined with fundamental scientific understanding of gas capture in the porous host, the solvent, and the resulting PL to enable material design of tunable PL compositions.

■ GAS ADSORPTION PROPERTIES

In order to tune the large number of possible PL compositions for gas adsorption applications, it will be necessary to predict the adsorption properties of PLs prior to their synthesis. Understanding the relationships between PL structure and adsorption requires detailed and fundamental understanding of adsorption capacity, binding mechanisms, and selectivity in both the individual components of the PL and the complete material system. In the following sections we will outline the current understanding of the mechanisms and structure–property relationships that result in improved gas adsorption properties in PLs, along with challenges and opportunities for future study.

1. Adsorption Capacity. PLs have consistently reported increased gas adsorption compared with the neat liquid or the solid porous host. Multiple mechanisms have been proposed for this phenomenon but vary with PL composition. For example, it is assumed that the gas adsorption capacity will increase linearly with the amount of dissolved porous hosts, as the dissolved gas should be preferentially adsorbed within the introduced porosity. This assumption holds for a few Type 3 PLs compositions, including ZIF-8 or HKUST-1 in ionic liquids and ZIF-8 in silicone oil, olive oil, or polyethylene glycol as the solvent.^{22,23,31}

Alternatively, some Type 2 PLs exhibit significant increases in CO₂ and CH₄ capture with even small additions of a dissolved porous host. For a Type 2 PL formed from an imine cage and an ionic liquid, an 8× increase in CH₄ adsorption was reported with only 0.7 volume percent dissolved hosts.¹⁵ Similarly, the introduction of vicinal diamines to a Type 2 PLs with a CC13 cage and a crown-ether solvent had twice the CH₄ adsorption capacity as a CC13 + ionic liquid Type 2 PL *without* the vicinal diamines.²⁸ Therefore, the gas molecules must be adsorbing not just inside the pore space, but elsewhere in the system. This is supported by recent data from Shan et al.²² They developed a Type 3 PL based on ZIF-8 + ionic liquid ([DBU-PEF][NTf₂]) with a CO₂ adsorption capacity of 1.56 mmol/g with 30 wt % nanoparticles that is 2–3× higher than Type 3 PLs based on the zeolites ZSM-5 or Silicalite-1.²² The improvement in the CO₂ adsorption capacity is attributed

to the higher surface area of the ZIF-8 nanoparticle and preferential diffusion of CO₂ into the ionic liquid indicating that the past focus on the amount of porous host and the pore size/pore window may not capture all the mechanisms available to develop a high CO₂ capacity PL composition.

Clearly, further evaluation of where and how the different gas molecules are adsorbed in the PL is necessary, including how CO₂ reacts with the solvent, the porous host, and at the host–solvent interface. For example, the introduction of a dissolved host may introduce binding sites at the porous host–solvent interface rather than primarily capturing the gas molecule inside the pore. Alternatively, the presence of the porous host may stabilize the cavities within the solvent itself, providing the solvent rather than the porous host with a larger adsorption capacity. Molecular scale modeling (gas phase electronic structure modeling, density functional theory (DFT), classical molecular dynamics (MD)) can provide the opportunity to evaluate these gas binding mechanisms and their role in the nonadditive effects on gas adsorption capacity seen in the Type 2 PLs.¹⁵ These effects have also been postulated as the key to understanding the reported low-temperature regenerative properties of PLs,^{23,54} one of the potential benefits of PLs over existing liquid carbon capture systems.

2. Gas Binding Mechanisms. The high gas adsorption in PLs has been attributed to the existence of multiple gas binding sites and have been identified both computationally and experimentally. Recently, Fourier transform infrared spectroscopy (FTIR) identified multiple CO₂ binding environments within a ZIF-8 + ionic liquid ([DBU-PEG][Tf₂N]₂) Type 3 PL, suggesting that a combination of free/physiosorbed CO₂ was the cause of the reported high CO₂ adsorption capacity.³⁶ However, the complexity of the PL structure, including the multiple interfaces (solvent–air, solvent–solvent, solvent–host) complicate the ability to characterize the system. Therefore, computational methods such as DFT and classical MD simulations provide the atomic scale resolution necessary to identify different binding sites and their relative stability in PL systems. For example, six unique binding sites were identified via DFT simulations of CO₂ binding in a POC + crown-ether Type 2 PL, including inside the pore, on the surface of the porous host, in the pore widow, along the vertex of the POC, between the POC and the solvent, and between two solvent molecules.⁵⁵ Figure 3 includes snapshots of these representative CO₂ binding sites. The distribution of binding sites and their relative stability also factors into the adsorption capacity of different gas molecules. For instance, dissolved CH₄ gas molecules are found to be more stable in the branched functional groups in a dissolved crown-ether-substituted POC molecules in a 15-crown-5 solvent that formed a Type 2 PL, while the smaller CO₂ was primarily captured by the cage structure itself.⁵⁶

Overall, molecular modeling evaluation of gas–PL interactions have identified physisorption-based interactions, but mechanisms of gas capture are specific to each PL composition. This complicates identification of mechanisms on which to base PL design mechanisms. Molecular scale computational modeling of limited PL compositions has been used to identify the specific mechanisms of gas capture, primarily for CO₂ adsorption in Type 2 PLs. For example, an extensive evaluation by Yin et al.³⁰ identified that CO₂ is particularly stable in the pore window of POCs in a Type 2 PL composed of a POCs and hexachloropropene; this is due to

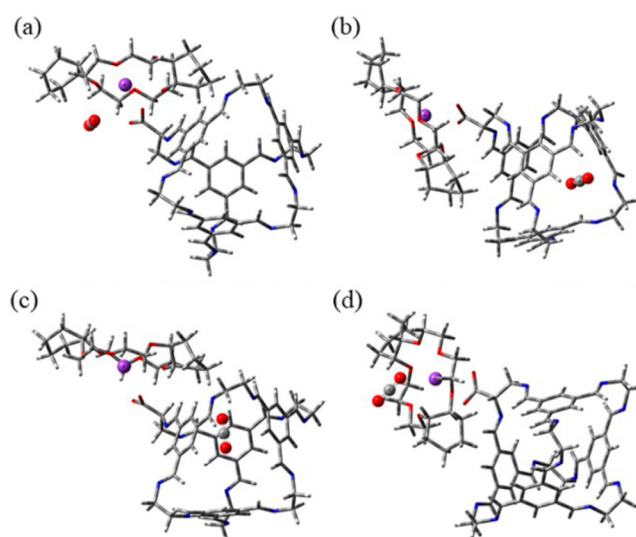


Figure 3. Optimized configurations for 18-C-6 PL adsorbing CO₂ lies (a) between the crown and the cage, (b) at the widow, (c) inside the cage, and (d) outside the cation. Reproduced with permission from ref 55 (2021, Elsevier).

hydrogen bonding between the oxygen in the CO₂ molecule and hydrogen atoms in the pore window. When comparing CO₂ versus CH₄ capture, the strong π – π interaction between the CO₂ and the benzene ring result in preferential binding compared with weaker C–H/Cl $\cdots$ π interactions between CH₄ and the benzene ring. Hydrogen bonding was also identified as the primary bonding type between either CO₂ or SO₂ and a hollow silica-based Type 1 PL.^{57,58} Finally, hydrogen bonding as a gas adsorption mechanism is not exclusively between the porous host and the gas molecule. CO₂ was primarily adsorbed by the anionic component of the [P_{6,6,6,14}][NTf₂]₂ ionic liquid used in a Type 3 PL with ZIF-8 as the porous host.³³

A comprehensive and fundamental understanding of the location and distribution of gas binding sites, including competitive binding between different gas molecules, will be necessary for the development of material design principles for PLs. Thus, far, molecular modeling approaches have identified that gas binding energies in PLs are through weak physisorption interactions that vary with the PL composition. The use of ionic liquids has complicated analysis of physisorption mechanism since they create a four-component PL composition (porous host, gases, and the anionic and cationic ionic liquid components) that requires a fundamental understanding of gas binding properties between each constituent. While the interactions between dissolved gases and ionic liquids have been evaluated for varying gas capture applications,^{59–61} it is not clear how the properties of the ionic liquid would be altered in the presence of a dissolved porous host.

Here, DFT molecular scale modeling can be leveraged to identify and isolate binding effects for evaluation of universal binding mechanisms as a function of the composition of the solvent or the porous host. For instance, if the same gas binding mechanisms occur between the gas and the ionic liquid, regardless of the presence of a porous host, then rapid material design could occur without requiring evaluation of every ionic liquid and porous host combination.

3. Gas Selectivity. PLs not only have reported higher gas adsorption capacity than the neat liquid or the solid porous

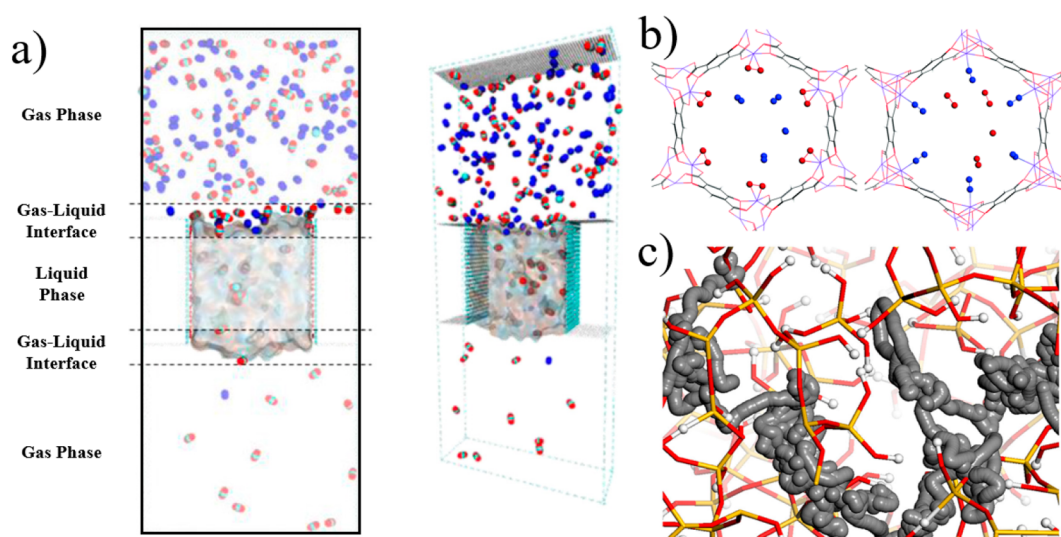


Figure 4. (a) MD modeling of CO₂ (red–light-blue) and N₂ (dark blue) separation by a membrane supported ionic liquid at room temperature, reproduced with permission from ref 80 (Elsevier, 2022), (b) direct competitive N₂/O₂ separation in a M₂(DOBDC) MOF framework via AIMD simulations, reproduced with permission from ref 76 (Royal Society of Chemistry 2016), and (c) reactive MD simulation of the diffusion pathway of water in a nanoporous silica framework, reproduced with permission from ref 88 (Springer Nature, 2018).

host, but also differences in selectivity. Early investigations indicated that the PLs mimic the shape- and size- selectivity of the POC that serves as the porous host in Type 2 PLs.^{15,62,63} Therefore, porous hosts with known size selectivity were combined with sterically excluded solvents for the design of new PLs. One example is in the selective capture of ethene over ethane in a Type 3 PL composed of an AgA zeolite in an ionic liquid.⁵⁴ This approach is particularly effective when gas binding is based on size selectivity of the pore window. For instance, narrowing the cage window of a scrambled POC (CC3³:13³-R) Type 2 PL effectively stopped Xe gas uptake.⁶⁴

In addition to the size selectivity, the binding energy of different molecules in the porous host structure also influences gas selectivity. In a crown-ether-substituted cage structure, the binding energy for the gas molecule in the center of the porous host was strongest for CO₂ (−22.2 kJ/mol), followed by CH₄ (−17.3 kJ/mol), and then N₂ (−13.3 kJ/mol).⁶⁵ Therefore, the selectivity of the CO₂ in the cages was caused by stronger binding in the pore, strengthening the argument that properties of the porous host control gas selectivity during adsorption.

Yet, the entirety of the improvements in selectivity in PLs compared to porous materials cannot be attributed to the size and shape selectivity of the porous host alone. For example, a Type 3 PL formed from a ZIF-8/glycol-2-methylimidazole slurry exhibited an order of magnitude difference in CO₂ selectivities of ~950 CO₂/H₂, ~400 CO₂/N₂, and ~140 CO₂/CH₄ compared to the solid ZIF-8.^{66–68} The high selectivity was attributed to the adsorption of CO₂ in glycol-2-methylimidazole solution, so that the distribution of gas molecules available for capture in the pore space is different than that of the source.⁶⁹

Another possible explanation is that the presence of the porous hosts is changing the structure of the solvent. Higher concentration of porous hosts has been reported to increase CO₂ selectivity in a Type 2 crown-ether substituted POC dissolved in a 15-crown-5 solvent. This occurs because the presence of pore channels allows CO₂ to diffuse more efficiently through the PL.²⁸ The ability of the solvent to

alter gas adsorption selectivity is not well studied, with most of the current research focuses on the concentration and functionalization of the porous hosts, leaving a clear opportunity in tuning PLs for gas adsorption.

If PLs are to be used in highly selective applications, a detailed understanding of the varying sources of selectivity in PLs will need to be identified mechanistically. Currently it is clear that multiple features of the PL control the gas adsorption selectivity, including the size and shape selectivity of the cage, the binding energies between the gas molecule and the porous host, and the concentration of the gas molecules in the solvent. These effects have been evaluated and identified for different gas species (CO₂, CH₄, N₂, Xe) in individual Type 2 and Type 3 PLs. It is unclear how these theories and observations apply across the spectrum of PL compositions. Evaluation of the transferability of the identified gas selectivity mechanisms between PL compositions will allow for transferability of the results for PL design.

Additionally, further questions remain on the sources of gas selectivity. For instance, there has been limited evaluation of how gas diffusion within solvent impacts the gas capture selectivity. The presence of the dissolved host itself may be perturbing the structure of the solvent, forming secondary pockets of porosity or enabling faster gas diffusion that may increase gas selectivity. Some features of this have been reported by Greenaway et al.²⁸ who identified chiral selectivity for adsorption of chiral aromatic alcohols. They attributed this to the loss of solvent organization in the presence of the porous host. Further use of computational methods, such as classical MD simulations of the ionic liquid and porous host combinations, can be used to identify changes in medium range order in the solvent due to the presence of high concentrations of the porous host.

FUTURE NEEDS AND RESEARCH DIRECTIONS

PLs are poised to provide transformation capabilities in gas adsorption, based on their high tunability from multiple gas binding and separation mechanisms. We see unique opportunities for the future of PLs in the creation of highly selective

gas adsorption systems, particularly in materials design for direct air capture (DAC) of CO₂. Since CO₂ has a concentration of only ~400 ppm in the atmosphere, DAC requires extremely high selective adsorption of CO₂ over N₂ and O₂ which is not possible in current porous materials or liquid capture systems.^{70–72} PLs can answer these challenges through the amalgamation of multiple mechanisms for increasing selectivity. For instance, a combination of size and shape selectivity in the porous host, multiple strong binding sites in the cage and within the solvent, and selective diffusion of CO₂ into the bulk solvent has the potential to form a highly selective gas capture system within a single PL composition. Therefore, this field is ripe for the exploration of the rules governing these combined mechanisms.

Creation and design of a PL with the highly tuned gas capture properties requires fundamental and mechanistic understanding of gas adsorption, binding, and selectivity that can be transformed through the application of computational methods. Significant work has already been initiated on the use of computational methods for analysis of PLs, including DFT^{30,55,57,58,73} and classical MD simulations^{15,28,69,73,74} for identification of binding sites and energies. A challenge here is the ability to model increasingly complex PL compositions with sufficient fidelity to identify the mechanisms of gas capture. We feel that three main factors need to be considered for material design of highly selective PL compositions: (i) evaluation of competitive binding mechanisms, (ii) thermal and vibration effects on the stability of binding sites, and (iii) nanoconfinement effects on the properties of the solvent phase.

First, it will be critical to apply computational methodologies such as *ab initio* molecular dynamics (AIMD), that account for direct competitive effects in calculation of gas binding sites. Static calculations can provide the basis for identifying stable binding sites in individual PL components, but thermal or steric factors can preclude the ability of a gas molecule to access a particular binding site and influences its stability. AIMD uses finite-temperature dynamic trajectories with forces that are computed from electronic structure calculations, which allows for simulation of chemical processes from condensed phases without requiring a user designed force field.⁷⁵ The same methods have been applied for evaluation of gas separation in other porous MOF materials, see snapshots of O₂ and N₂ separation in a M₂(DOBDC) MOF framework in Figure 4b.^{76,77} We have also reported that competitive NO₂ and H₂O adsorption in rare-earth DOBDC MOF structures is modeled via AIMD, the NO₂ is clearly bound to the linker rather than the metal center, a significant change from what was assumed via static calculations.⁷⁸ Similarly, in PLs, a comparison of the binding energies inside and outside the porous host is ultimately insufficient to capture the competitive adsorption effects that influence the ability of the porous host to capture the gas molecule. As an example, a binding site in the pore window⁵⁵ may block the pore from being filled by more than one gas molecule, limiting gas adsorption capacities. Furthermore, the faster diffusion of a lighter molecule (e.g., H₂O) to a semistable binding site might block a CO₂ binding site. High fidelity and dynamic DFT modeling such as AIMD could be used to identify the binding energies and dynamic reactions that occur during gas adsorption in a methodology that is transferable to different PL systems.

Second, current DFT simulations do not consider the temperature and vibrational effects that can alter the stability of gas binding sites. Even at low temperatures, variation in pore

window sizes can alter the ability of a gas molecule to diffuse into a pore as we have seen in the evaluation of H₂ and D₂ diffusion into and out of RCC3, CC3-S, and 6ET-RCC3 POCs.⁷⁹ The functionalization of the porous host and the solvent molecule can interfere with the binding geometries of weakly bound gas molecules. Additionally, the relatively weak physisorption of the gas molecules in PLs,^{30,31} combined with their low energy regeneration procedures,^{23,54} indicate that this may be of specific concern in these materials. Dynamically evolving systems at service temperatures (300–500 K) will be required to evaluate thermal effects on gas adsorption capacities and selectivities. For example, MD simulation of the separation of CO₂ (red-light blue) and N₂ (dark blue) by a membrane supported ionic liquid has been evaluated at room temperature by Fang et al. in Figure 4b.⁸⁰ Classical MD simulations have excellent applicability in the evaluation of thermal effects using existing force fields, such as OPLS-AA⁸¹ and COMPASS⁸² force fields along with ionic liquids^{83,84} and MOF/POC^{85,86} force fields. For increasingly complex PL compositions, force field development will need to be undertaken, potentially using DFT data generated from evaluation of competitive gas binding simulations to enable evaluation of these thermal effects.

Finally, the high concentration of porous hosts in the PL can introduce nanoconfinement effects in the surrounding solvent molecules. Porous host concentrations can be as high as 30–40 wt % in PLs. Therefore, the properties of the nanoconfined solvent are different than when the solvent is a bulk fluid. We have seen this in the simpler case of aqueous solvents in both porous silica frameworks and zeolites,^{87,88} where the diffusion rates are slowed and intermolecular structure becomes more structured under confinement. In particular, nanoconfinement can limit diffusion of dissolved ions and gas molecules, as seen in a water molecule diffusion pathway in a nanoporous silica gel structure in Figure 4c.^{89,90} In PLs, gases need to diffuse through the solvent to enter the porous hosts for adsorption, a process that could be substantially hindered in a nanoconfined solvent. Alternatively, nanoconfinement is associated with incomplete packing of solvent molecules⁹¹ so that a lower solvent density occurs at the solvent–host interface. This space may serve as an opportunity for either faster gas diffusion or stable binding sites in the PLs.

Current evaluation of ionic liquids nanoconfinement focuses on the structure in relationship to an atomically flat surface;⁹² the evaluation of ionic liquids confined between nanoporous hosts with complex functionalized surfaces is poorly characterized. There has been some additional focus on the use of computational methods (classical MD and DFT) for predicting the rheological and mass transport properties of ionic liquids and liquid mixtures of polymer solutions.^{93–97} These capabilities could be leveraged for evaluation of PLs prior to synthesis, to identify structural features that control the liquid properties of the system. Ultimately, advanced molecular scale simulations, either larger scale classical MD or AIMD simulations using DFT can be leveraged to answer these critical questions on the role of solvent–host nanoconfinement in altering the gas adsorption properties.

CONCLUSIONS

PLs clearly demonstrate unique gas adsorption capabilities based on the existence of multiple gas binding sites, weak physisorption mechanisms, and changing gas diffusion and stability in the solvent. However, directed research into the

complex mechanisms governing targeted species is still a new field. Herein, we have discussed the four types of PLs, with the Type 2 and Type 3 PL compositions exhibiting the most potential for material design and tunability based on the composition of the solvent and the porous host. The immense number of possible PL composition highlights the clear need for complex computational approaches to materials development without relying on evaluation of individual PL compositions. By targeting fundamental understanding of the properties of PL that result in high adsorption capacity and selectivity, targeted material design of new PL compositions with system specific gas adsorption and separation properties can be performed.

To achieve these goals of material design of PLs for gas adsorption, we propose the use of computational methods such as AIMD for evaluation of gas adsorption mechanisms in PLs. This will allow us to design PLs with specific structural features necessary to tune (i) the direct competitive gas binding mechanisms, (ii) the thermal effects on the stability of the gas binding sites, and (iii) the influence of nanoconfinement of the solvent on the gas diffusion and adsorption. This progresses beyond evaluation of static binding energies at low temperatures, which has provided a strong initial basis for understanding the complexity of PL systems, to the inclusion of more complex and realistic gas binding mechanisms. In particular, the physisorption mechanisms for gas adsorption are likely to be sensitive to competitive binding and thermal effects, highlighting that they need to be evaluated directly. With the new insight that can be gained from molecular scale modeling studies, our current understanding of PLs can be transformed enabling current PL compositions to be only the beginning of a new material class with tunable properties for targeted gas adsorption and separation.

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Notes

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NOMENCLATURE

[DBU-PEG][Tf₂N]₂ = 8,8'-(3,6-dioxaoctane-1,8-diyl)bis-(1,8-diazabicyclo[5.4.0]undec-7-en-8-ium) bis-(trifluoromethanesulfonyl)imide
[P_{6,6,6,14}][NTf₂] = trihexyltetradecylphosphonium bis-(trifluoromethylsulfonyl)imide
TOMAC = trioctylme-thylammonium chloride
DOBDC = 2,5-dioxido-1,4-benzenedicarboxylate

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